PORTFOLIO MANAGEMENT

Work Project # 2

Scenario

Portfolio management is going through a renaissance. Why? Machine learning is changing the way people make decisions under uncertainty. Relying on time-tested, Nobel-Prize winning ideas sounds good, but in fact, these ideas may be outdated. Machine learning has already changed the way statistical analyses are done. When applied to the portfolio optimization process, that improvement could be enough to provide an advantage. Combined with leverage and other factors, these advantages could mean a substantial and significant outperformance over the competition.

Welcome to the 21st century of portfolio management! In this GWP, students will apply machine learning concepts of denoising, clustering, and backtesting to optimize a portfolio. Use the portfolios that were analyzed in the previous GWP.

Tasks

<u>Choose portfolios of at least 10 assets.</u> Use January 1, 2024 - September 30, 2025 as the reference (training) period and use daily data.

Step 1. Determine the optimal Kelly portfolio. Can you find the optimal Kelly portfolio if no asset is allowed to be more than 20% of the total portfolio. If you can, find the optimal portfolio with such a constraint in place.

Step 2. Using data from October 1, 2025 to December 31, 2025, compare the Kelly portfolio with the double-Kelly and double-Kelly portfolio in terms of performance and risk (cumulative return, Sharpe ratio, Maximum Drawdown, 95%-cVaR). Organize your findings in a nice table so that the performance and risk of each are showing clearly. **NOTE**: *The model is to be estimated/trained using the data from January 1, 2024 to September 30, 2025 (in-sample period) and tested on the period October 1, 2025 to December 31, 2025 (out-of-sample period).*

Step 3. How would you backtest the model(s) in Step 1 and Step 2 using K-fold cross validation. Explain the process in half a page or so, then write the Python code to perform the backtesting and report the findings. (Use the entire 2 years of data from January 1 2024 to December 31, 2025 to do this Step).

Step 4. Then each student should explain (in no more than 2 pages per item) the features and benefits of each of these topics:

- Improvements using denoising
- Improvements using clustering. Ideally here you should discuss HRP (Hierarchical Risk Parity). If you use a different method, be sure to explain what you did. For example, groups could also opt for an easier to implement HEW (Hierarchical Equal Weight) portfolio or another method of your choice.
- Improvements using detoning.

Step 5. Students work together to apply the methodologies described in Step 4 to the portfolio used in Step 1 above. You can use third party software, but make sure that you explain the steps of what you are doing. In addition, students should prepare a document explaining the merits of applying one, two, or all of these “improvements” to the current portfolio. They could also theorize about what could be the best sequence with which to use the three tools listed above. (Be essential, do not “write a book”!)

Step 6. Students discuss if and how multiple improvements potentially outperform single improvements. Students show which combination of multiple improvements (if any) outperforms single improvements, according to at least three metrics of the group’s choice. Examples of metrics include return, Sharpe ratio, expected shortfall, variance, maximum drawdown, Sortino ratio, etc.)

Discuss if the ML methodologies helped the out of sample performance of the portfolio vs the more traditional approach using (some of the) metrics described in Step 4. You should do this by considering the following:

1. What are the differences in performance according to these metrics for the different combination of improvements?
2. What reasons might explain these differences in performances (with specific reference to the financial products, the historical data, the math of the model dynamics, etc.),
3. In which cases, if any, does the incremental gain in performance according to the chosen metrics justify the additional complexity and effort of the improvements?

Step 7: Preparing for a Technical Talk

1. Prepare 3 – 5 slides on a technical talk about how the portfolio methodology addressed and solved problems in 1 of the following 3 areas:
    a. Regularization
    b. Prediction vs. Interpretation
    c. Estimation Error
    d. Regime Shifts
2. Be sure to include definitions, equations, formulas about the term
3. Show a direct connection between the data you analyzed and the challenge selected.
4. Show a direct connection between the method you used and how it addressed the financial challenge (Note: Methods may not necessarily help solve the problems caused by the specific challenge)